Problem 17.1.11

Sketch or graph the given region and its image under the given mapping.

$$|z| \leqq \frac{1}{2}, \quad -\frac{\pi}{8} < \operatorname{Arg} z < \frac{\pi}{8}, \quad w = z^2$$

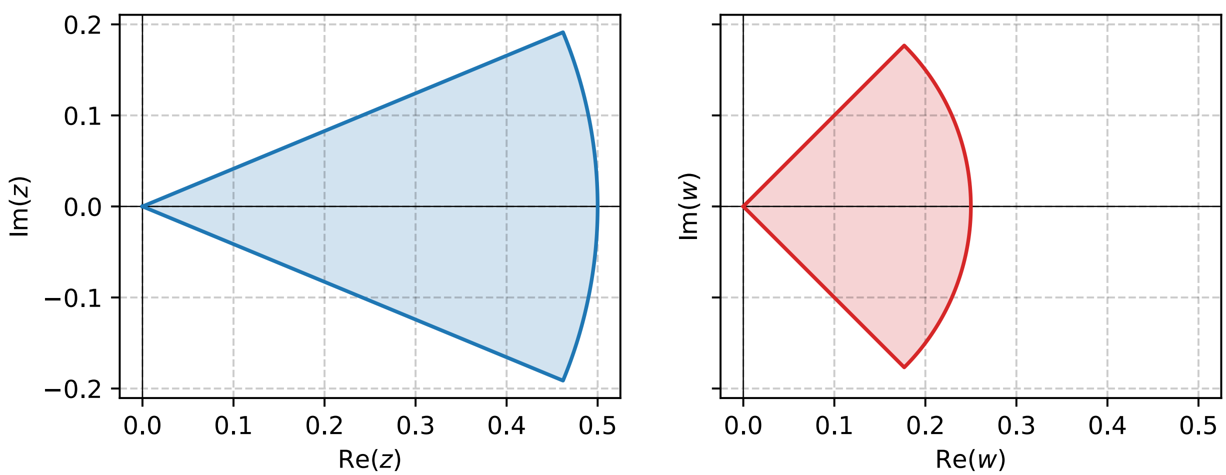
Solution

$$z = re^{i\theta}$$

$$w = Re^{i\phi}$$
$$z^2 = r^2 e^{2i\theta}$$

$$R = r^2 = \frac{1}{4}$$

$$\phi = 2\theta \implies -\frac{\pi}{4} < \operatorname{Arg} w < \frac{\pi}{4}$$



Problem 17.1.13

Sketch or graph the given region and its image under the given mapping.

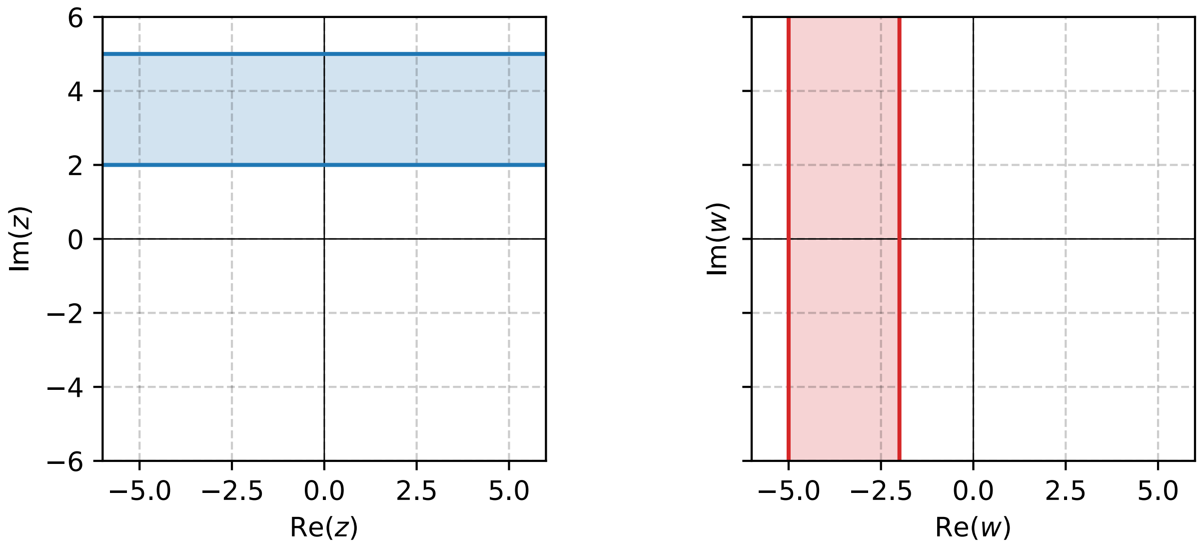
$$2 \leqq \operatorname{Im} z \leqq 5, \quad w = iz$$

Solution

The region is an infinite strip bounded by $z_1 = 2i$ and $z_2 = 5i$, mapping to $w = iz$ gives another infinite strip bounded by

$$w_1 = iz_1 = -2$$

$$w_2 = iz_2 = -5$$



Problem 17.1.15

Sketch or graph the given region and its image under the given mapping.

$$\left|z - \frac{1}{2}\right| \leqq \frac{1}{2}, \quad w = \frac{1}{z}$$

Solution

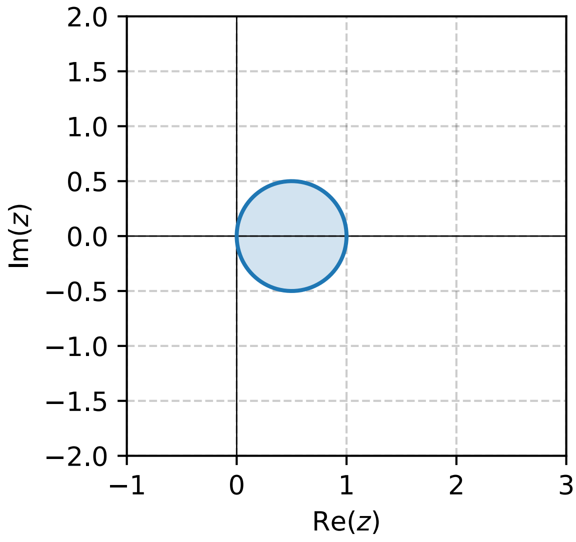
The boundary of the region $z = x + iy$ has the equation

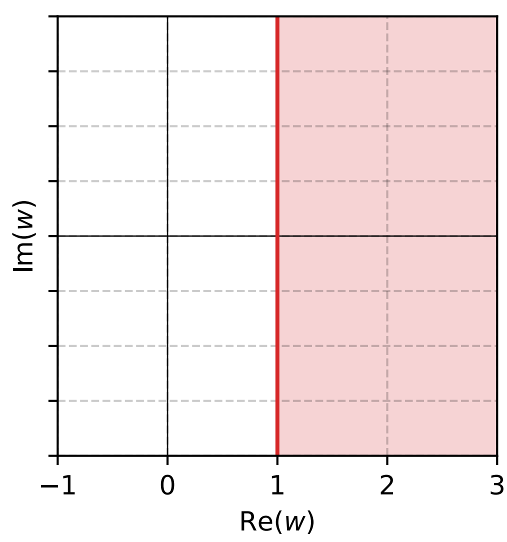
$$\begin{aligned}
\left(x - \frac{1}{2}\right)^2 + y^2 &= \left(\frac{1}{2}\right)^2 \\
x^2 - x + \frac{1}{4} + y^2 &= \frac{1}{4} \\
x^2 - x + y^2 &= 0 \\
x^2 + y^2 &= x \implies |z|^2 = \operatorname{Re} z
\end{aligned}$$

Applying this relation to $w = u + iv = \frac{1}{z}$ we can find the boundary of the mapping.

$$\begin{aligned}
\left|\frac{1}{w}\right|^2 &= \operatorname{Re}\frac{1}{w} \\
\frac{1}{|w|^2} &= \operatorname{Re}\frac{1}{u+iv} \\
\frac{1}{|w|^2} &= \operatorname{Re}\frac{u-iv}{u^2+v^2} \\
\frac{1}{|w|^2} &= \frac{u}{|w|^2} \\
u &= 1
\end{aligned}$$

This is a vertical line. Since $z_c = \frac{1}{2}$ is within the original region, then $w_c = \frac{1}{z_c} = 2$ which is on the right side of the boundary and therefore the mapped region is the half plane on the positive side of $u = 1$





Problem 17.1.17

Sketch or graph the given region and its image under the given mapping.

$$\operatorname{Ln} 2 \leqq x \leqq \operatorname{Ln} 4, \quad w = e^z$$

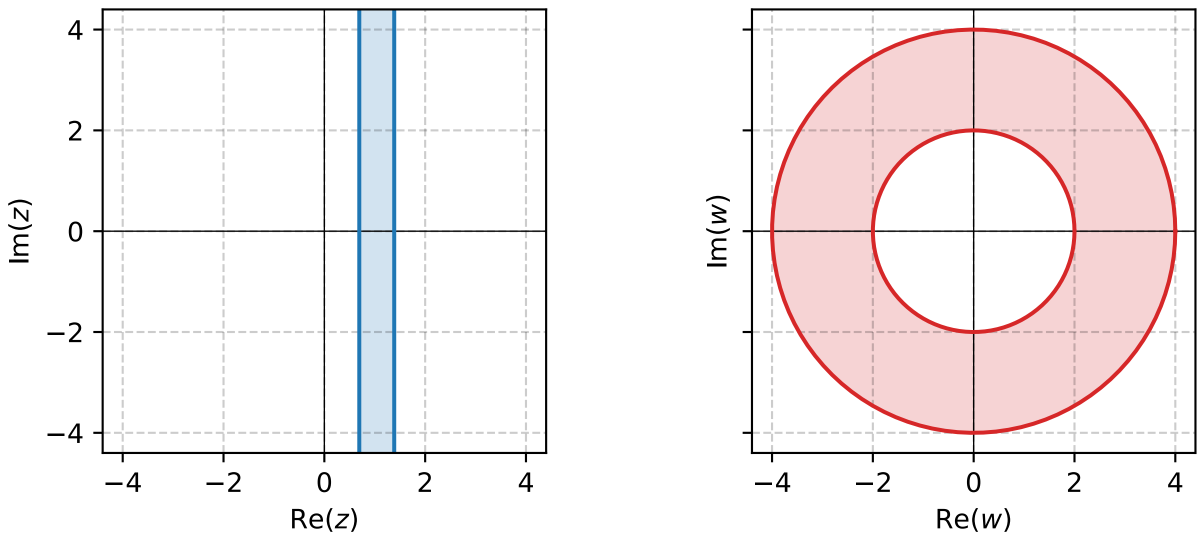
Solution

The region is an infinite vertical strip bounded by $z_1 = \operatorname{Ln} 2$ and $z_2 = \operatorname{Ln} 4$,

$$|w_1| = |e^{z_1}| = \left|e^{\operatorname{Ln} 2}\right| = 2$$

$$|w_2| = |e^{z_2}| = \left|e^{\operatorname{Ln} 4}\right| = 4$$

The new boundary is two concentric circles.



Problem 17.1.19

Sketch or graph the given region and its image under the given mapping.

$$1 < |z| < 4, \quad \frac{\pi}{4} < \theta \leqq \frac{3\pi}{4}, \quad w = \operatorname{Ln} z$$

Solution

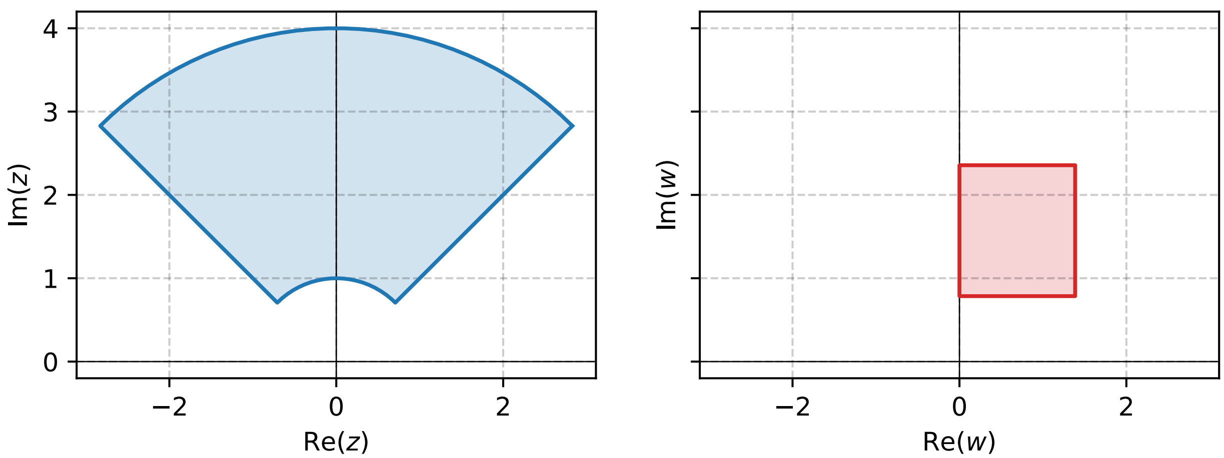
The region is bounded by 4 borders that can be described in the form $z = re^{i\theta}$.

Mapping to

$$\begin{aligned} w &= \operatorname{Ln} z \\ &= \ln|z| + i \operatorname{Arg} z \\ &= \ln r + i\theta \\ &= u + iv \end{aligned}$$

$$u = r \implies \ln 1 < u < \ln 4$$

$$v = \theta \implies \frac{\pi}{4} < v \leqq \frac{3\pi}{4}$$



Problem 17.2.7

Find the inverse $z = z(w)$ Check by solving $z(w)$ for w.

$$w = \frac{i}{2z - 1}$$

Solution

$$\begin{aligned} w &= \frac{i}{2z-1} \\ 2z - 1 &= \frac{i}{w} \\ z &= \frac{i}{2w} + \frac{1}{2} \end{aligned}$$

$$\boxed{z(w) = \frac{i}{2w} + \frac{1}{2}}$$

Problem 17.2.13

Find the fixed points.

$$w = 16z^5$$

Solution

$$\begin{aligned} f(z) &= z \\ w = 16z^5 &= z \\ 16z^5 - z &= 0 \\ z\left(16z^4 - 1\right) &= 0 \quad \Longrightarrow \boxed{z = 0,\ \pm\frac{1}{2},\ \pm\frac{1}{2}i} \end{aligned}$$

Problem 17.2.15

Find the fixed points.

$$w = \frac{iz+4}{2z-5i}$$

Solution

$$\begin{aligned}
f(z) &= z \\
w = \frac{iz+4}{2z-5i} &= z \\
iz+4 &= 2z^2 - 5iz \\
0 &= 2z^2 - 6iz - 4 \\
0 &= z^2 - 3iz - 2 \qquad \Longrightarrow \boxed{z = \frac{3i \pm i}{2} = i, 2i}
\end{aligned}$$

Problem 17.2.19

Find all Linear Fractional Transformations (LFTs) with fixed point(s).

$$z = \pm i$$

Solution

$$w = \frac{az+b}{cz+d}$$

$$w = z \implies cz^2 - (a-d)z - b = 0$$

$$z = i: \quad ci^2 - (a-d)i - b = -(c+b) - (a-d)i = 0$$

$$z = -i: \quad c(-i)^2 - (a-d)(-i) - b = -(c+b) + (a-d)i = 0$$

From the real parts: $c = -b$

From the imaginary parts: $a = d$

$$\boxed{w = \frac{az+b}{a-bz}, \quad \forall a, b \in \mathbb{C}}$$

Problem 17.3.9

Find the LFTs that maps the given three points onto the three given points in the respective order.

$$1,\ i,\ -1 \text{ onto } i,\ -1,\ -i$$

Solution

$$\frac{w-w_1}{w-w_3}\cdot\frac{w_2-w_3}{w_2-w_1}=\frac{z-z_1}{z-z_3}\cdot\frac{z_2-z_3}{z_2-z_1}$$
$$\frac{w-i}{w-(-i)}\cdot\frac{(-1)-(-i)}{(-1)-i}=\frac{z-1}{z-(-1)}\cdot\frac{i-(-1)}{i-1}$$
$$\frac{w-i}{w+i}\cdot\frac{1-i}{1+i}=\frac{z-1}{z+1}\cdot\frac{i+1}{i-1}$$

$$\frac{1-i}{1+i}=\frac{(1-i)^2}{1^2+i^2}=\frac{-2i}{2}=-i$$

$$\frac{i+1}{i-1}=\frac{(i+1)^2}{i^2-1^2}=\frac{2i}{-2}=-i$$

$$\frac{w-i}{w+i}=\frac{z-1}{z+1}$$
$$(w-i)(z+1)=(z-1)(w+i)$$
$$wz+w-iz-i=zw+iz-w-i$$
$$2w=2iz$$
$$w=iz$$

$$\boxed{w=iz}$$

Problem 17.3.11

Find the LFT that maps the given three points onto the three given points in the respective order.

$$-1,\ 0,\ 1 \text{ onto } -i,\ -1,\ i$$

Solution

$$\frac{w-w_1}{w-w_3}\cdot\frac{w_2-w_3}{w_2-w_1}=\frac{z-z_1}{z-z_3}\cdot\frac{z_2-z_3}{z_2-z_1}$$
$$\frac{w-(-i)}{w-i}\cdot\frac{(-1)-i}{(-1)-(-i)}=\frac{z-(-1)}{z-1}\cdot\frac{0-1}{0-(-1)}$$
$$\frac{w+i}{w-i}\cdot\frac{-1-i}{-1+i}=\frac{z+1}{z-1}\cdot\frac{1}{-1}$$

$$\frac{-1-i}{-1+i}=\frac{(-1-i)^2}{(-1)^2+i^2}=\frac{2i}{2}=i$$

$$\begin{aligned}
\frac{w+i}{w-i}\cdot i &= \frac{z+1}{1-z}\\
(w+i)(1-z)\cdot i &= (z+1)(w-i)\\
(w-wz+i-iz)\cdot i &= zw-iz+w-i\\
iw-iwz-1+z &= zw-iz+w-i\\
iw-iwz-w-zw &= -iz-i+1-z\\
w(i-iz-1-z) &= -iz-i+1-z\\
w &= \frac{-iz-i+1-z}{i-iz-1-z}
\end{aligned}$$

$$\boxed{w=\frac{-(1+i)z+(1-i)}{-(1+i)z+(i-1)}}$$

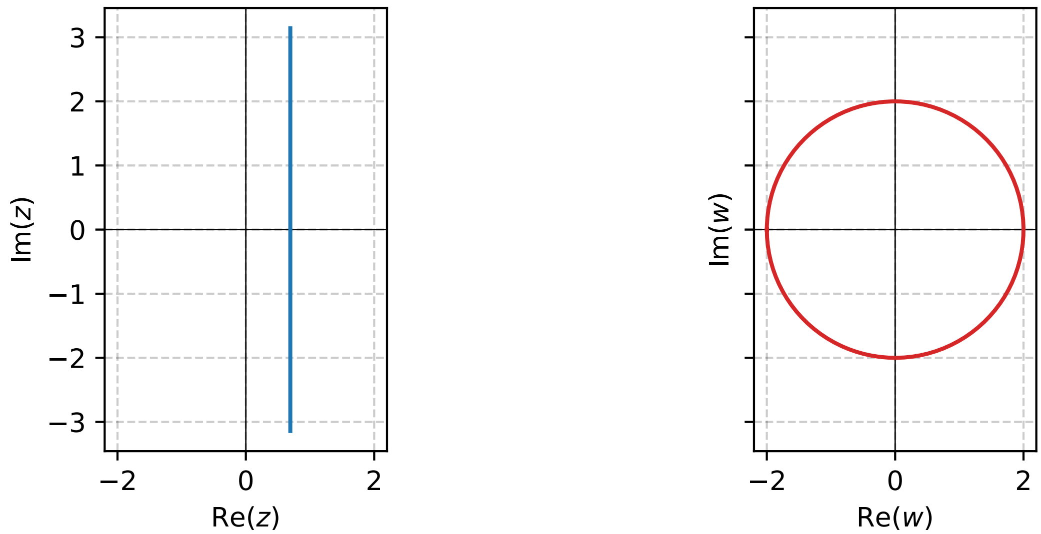
Problem 17.4.1

Find the image of $x = c = \text{const}$, $-\pi < y \leqq \pi$, under $w = e^z$.

Solution

$$z = x + iy = c + iy$$

$$w = e^c e^{iy}$$



Problem 17.4.3

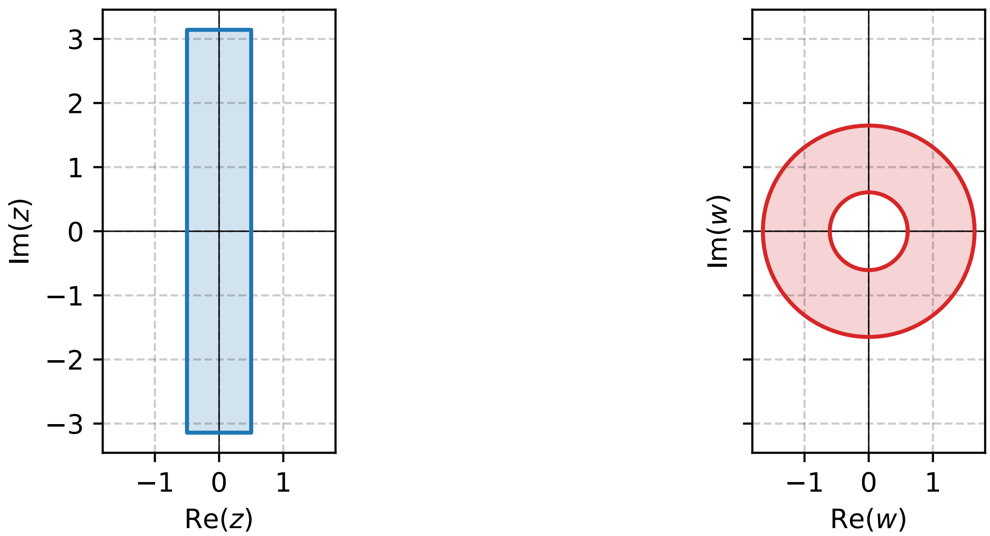
Find and sketch the image of the given region under $w = e^z$.

$$-\frac{1}{2} \leqq x \leqq \frac{1}{2}, \quad -\pi \leqq y \leqq \pi$$

Solution

$$z = x + iy$$

$$w = e^x e^{iy}$$



Problem 17.4.5

Find and sketch the image of the given region under $w = e^z$.

$$-\infty \leqq x \leqq \infty, \quad 0 \leqq y \leqq 2\pi$$

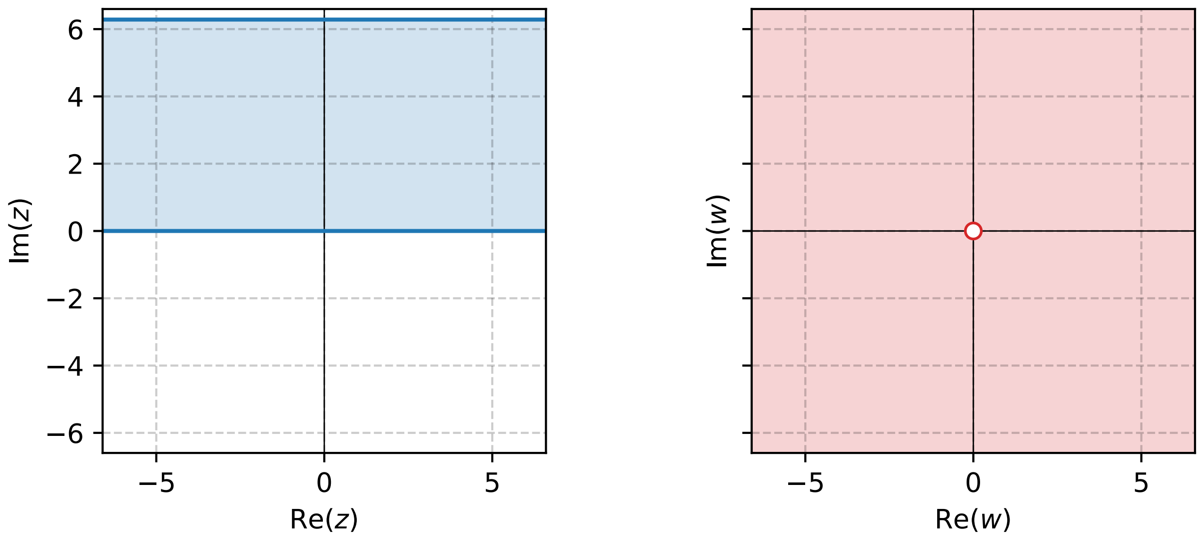
Solution

The initial region is an infinite strip between $0 \leqq y \leqq 2\pi$,

$$z = x + iy$$

The image fills the entire plane except at the center,

$$w = e^x e^{iy}$$



Problem 17.4.7

Find and sketch the image of the given region under $w = e^z$.

$$0 < x < 1, \quad 0 < y < \pi$$

Solution

Problem 17.4.11

Find and sketch or graph the image of the given region under $w = \sin z$.

$$0 < x < \frac{\pi}{2}, \quad 0 < y < 2$$

Solution

Problem 17.4.13

Find and sketch or graph the image of the given region under $w = \sin z$.

$$0 < x < 2\pi, \quad 1 < y < 3$$

Solution

Problem 17.4.19

Find the images of the lines mapping $x = c = \text{const}$ under the mapping $w = \cos z$.

Solution

Problem 17.4.21

Find and sketch or graph the image of the given region under the mapping $w = \cos z$ directly from **Problem 17.4.11**.

$$0 < x < \frac{\pi}{2}, \quad 0 < y < 2$$

Solution